Kaushal Vivek Abojwar

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kaushalvabojwar-gif.github.io/Portfolio1

SUMMARY

Control and automation engineer who programs PLCs for special purpose machines across Siemens, Allen-Bradley, Delta and Keyence platforms, and builds the data layer around them: sensor acquisition, HMI dashboards and ML-based fault detection. Built a predictive maintenance system selected for B&R's Open Automation Challenge 2026, and an IoT fault-detection system reaching 85–90% accuracy. Looking to develop and debug test and data acquisition systems as a System Engineer.

EXPERIENCE

Infinity Automations Systems Pvt Ltd 27 July 2026 – Present
Control Engineer Intern

- Program PLCs in ladder logic for special purpose machines (SPM).
- Worked on Siemens, Allen-Bradley, Delta and Keyence programming software across different machine projects.
- Worked on HMI development for machine operation and used LabVIEW alongside PLC programming.

ShreeTech Data Limited 06 Jan 2026 – 06 Feb 2026
Engineer Intern, Electrical / Switchgear

- Studied control systems and electrical control panels in an industrial switchgear environment.
- Observed motor-failure and downtime inefficiencies firsthand, which motivated the predictive maintenance project below.

EDUCATION

D.Y. Patil College of Engineering, Akurdi 2023 – 2027 (Expected)
B.E. in Instrumentation & Control Engineering, CGPA: 8.32/10 *Pune, Maharashtra*

Shree Maharishi Vidya Mandir, Chandrapur 2023
Class XII (CBSE), 79% *Chandrapur, Maharashtra*

Shree Maharishi Vidya Mandir, Chandrapur 2021
Class X (CBSE), 91.60% *Chandrapur, Maharashtra*

TECHNICAL SKILLS

Test & Data Acquisition: LabVIEW, sensor data acquisition, multi-sensor data processing, testing and debugging, technical documentation

PLC & HMI: Ladder logic, Structured Text, Siemens, Allen-Bradley, Delta and Keyence PLC software, B&R Automation Studio, mapp View HMI, SCADA, state machine design, motor control, alarm management

Embedded & IoT: Arduino, ESP32, MQ135/MQ3/MQ4/DHT22/FC-28 sensors, Blynk, Firebase, LPWAN, Wi-Fi

PROJECTS

PLC-Based Predictive Maintenance System with HMI Dashboard Team of 3
B&R Automation Studio, mapp View, Structured Text

- Designed and developed a predictive maintenance system for industrial motor control with an HMI dashboard built on B&R Automation Studio and mapp View; selected as a competing project in B&R's Open Automation Challenge 2026.
- Structured the control software using state machine design, with motor control, alarm management and role-based access control.
- Targeted unplanned motor failures in continuous manufacturing, where downtime can cost INR 10–15 lakhs per hour, by replacing reactive alarms with a predictive approach.

AI-Enabled IoT System for Real-Time Food Spoilage Detection Team of 5
ESP32, OpenCV, TensorFlow Lite, Blynk, Firebase

- Built real-time data acquisition from MQ135, MQ3, MQ4 and DHT22 sensors on an ESP32, streaming data to Blynk/Firebase for live monitoring and spoilage prediction.
- Developed an OpenCV and TensorFlow Lite module to detect visual spoilage cues, combined with pH data for classification.

Railway Infrastructure Monitoring System using IoT Sensors and Machine Learning Individual Project
LPWAN, Wi-Fi, SVM, RNN

- Engineered an IoT data acquisition system capturing vibration, temperature and pressure data over LPWAN and Wi-Fi networks.

- Processed multi-sensor data with SVM and RNN models for early fault detection (85–90% accuracy), enabling predictive maintenance with an estimated 20–25% reduction in unplanned downtime.

Arduino-Based Irrigation System with Real-Time Soil-Moisture Sensing Team of 4

Arduino UNO, FC-28 sensor, 12V pump

- Designed and built a threshold-based pump control system using soil-moisture sensor feedback to improve irrigation efficiency.
- Achieved about 18% water savings, reduced irrigation cycle time from 3.23 to 3.02 min, and validated performance at discharge rates of 7.67 and 7.42 ml/s.